Diagnose Before Solving: Lead-in Storyboard (Pre-Simulation)

v0.3, revised for the web build

Project owner: David Bacon
Date: October 23, 2026 (v0.1); revised November 11, 2026 (v0.2) and November 12, 2026 (v0.3)
Document status: Revised for publication
Based on: `claude/storyline-module-outline.md` (v0.3, six teaching slides)
Related: Scope Document 4.1, Decision Log Part 2 (D-165 onward) and Part 3 (D-173 onward)

Revision note (v0.2). The lead-in is now a static web page, not a Storyline module (D-173, D-177). The on-screen copy, correct answers, feedback text, and acronym rules are unchanged except where the web build needed different wording for an instruction (section 12 lists each change). Every "Storyline build" instruction from v0.1 is replaced by a "Build (web)" instruction. The filename keeps "storyline" so existing references work (D-186).

Revision note (v0.3). Brings the storyboard back into line with the published site after David's review of slides 3 and 4 (D-189): the slide 3 reveal buttons are relabelled, and slide 4 keeps the caller's problem on screen, lets the learner return to the explanation, and spaces its two defined terms correctly. The "Start the call" link now carries `?embed=1` (D-190). Section 12 lists every change.

How to read this document. Text in quotation blocks is on-screen copy. It matches the published site as of November 12, 2026; if the two ever differ, the site is what learners see. Everything else is a build description. Nothing here touches the simulation itself.

---

1. Global elements

Element	Specification
Layout	Responsive. One column on a phone, up to about 62rem wide on a desktop. Replaces the Storyline story size (1280 by 720), which no longer applies (D-186)
Navigation	Back and Next in a fixed bar at the bottom, with a progress indicator ("Slide 2 of 6"). Back is hidden on slide 1. On slides 2 to 5, Next appears only when the slide's activity is complete. On slides 1 and 6 the buttons inside the slide do the navigating
Addresses	Each slide has its own address (`#/1` to `#/6`), and the browser's Back button works
Glossary button	A persistent "Glossary" button in the page header opens a dialog (section 9). Built once, available on every slide
Skip button	Slide 1 only. Label: "Skip the lessons." Sub-label: "You will get a 30-second briefing first." Jumps to slide 6
Audio and video	None. Text-only, so no captions or transcripts are needed
Reading level	Plain language. Short sentences. Active voice
Acronym rule	Every acronym is spelled out on first use **on each slide where it appears**, because the skip button means no slide can assume an earlier one was read. The glossary is the backstop (section 10)
Feedback	Never color alone. Every correct or incorrect state also shows an icon and the words "Correct" or "Not quite"
Keyboard	Every interaction operable by keyboard. Practice cards are focusable and chosen with Enter or Space, then placed by choosing a group or slot. Dragging also works but is never required (D-181)
Screen readers	Feedback is announced as it appears; focus moves to each slide's heading when the slide changes; the check and cross icons are hidden from screen readers because the words "Correct" and "Not quite" carry the meaning
Images	None. The site uses no images, so no alt text is needed
Motion	The one small animation (a card shaking after a wrong placement) is switched off for visitors whose device asks for reduced motion
Tracking	None. Nothing is stored in the browser or sent anywhere (D-179). No slide is tracked for completion (D-162)

2. Slide 1: Your next call

Layout: Title top left. Short intro paragraph. A two-column area: the three numbered steps, and the objectives list. Next button, with the skip button beside it, below.

On-screen text:

> # Your next call
>
> You work in tech support for your own company. (Another name for this is IT support; **IT** stands for **information technology**.) Your callers are colleagues who depend on the tools you look after.
>
> Soon, a caller will phone in. He will not be happy.
>
> **How this works**
>
> 1. **Short lessons.** About eight minutes.
> 2. **A practice call.** You type, and he replies, like a live conversation.
> 3. **Detailed feedback** on how you handled it. You can try again.
>
> **By the end of the lessons, you will be able to:**
>
> - Open a tense call by acknowledging the person and what is at stake for them, before you start troubleshooting.
> - Ask open questions about what people experience when the problem happens.
> - Tell apart a software defect, a training gap, and a workflow mismatch.
> - Pass a problem to the person who can fix it, and promise only what you control.
>
> [Next] [Skip the lessons: You will get a 30-second briefing first.]

Interaction: buttons only.

Build (web):

- Next: go to slide 2.
- Skip: go to slide 6.
- "Troubleshooting" in the first objective is a link that opens the glossary at that term.

Design notes: the fourth objective deliberately avoids the word "escalate" on this slide to keep the opening light; escalation is introduced and defined on slide 5.

---

3. Slide 2: A symptom is not a cause

Layout: three parts on one slide, shown one at a time: (A) the principle, (B) the three kinds of cause, (C) the sorting activity. Parts A and B each have a Continue button. Next in the bottom bar stays hidden until part C is complete.

Part A: the principle

> # A symptom is not a cause
>
> A **symptom** is what someone notices: "The report is slow."
> A **cause** is why it happens.
>
> Fix a symptom without finding the cause, and three things follow:
>
> - **You lose time.** The problem comes back.

- **You lose trust.** The caller learns that you are guessing.
- **The real problem keeps going.** Nobody is working on it.

[Continue]

Part B: three kinds of cause

Three kinds of cause

Most problems you hear about fall into one of three groups.

Software defect. The software is not doing what it was built to do.
The clue: it fails the same way for everyone, even when it is used exactly as intended.

Training gap. The software works, and it fits the work, but the people struggling were never shown how to use it.
The clue: people who were shown it have no trouble. People who were not shown it do.

Workflow mismatch. The software works as built, but it does not fit how the work actually happens. (A **workflow** is the series of steps a person follows to get a piece of work done.)
The clue: the people affected know how to use it, and it still gets in their way at one particular point in the work.

[Continue]

Design note: the three cards have the same format and length, and the clues are written to be equally specific, so no category reads as the usual answer (D-152). The workflow clue is written to be general; it does not describe workarounds, wrong data, or sequencing, which would echo the simulation's scenario (D-151).

Part C: sort the cases

Sort the cases

Pick a case, then choose the kind of cause it points to. (You can also drag a case onto a group.)

Cards (display in this order) and correct groups:

#	Card text	Correct group
1	"Print jobs sent to the third-floor printer disappear. It happens to everyone, on every computer, no matter who sends them."	Software defect
2	"Three new hires cannot find where to enter their benefits choices in the **HR** (human resources) portal. All three started this week, and none was shown the portal. Employees who were shown it use it without trouble."	Training gap
3	"A clinic's check-in kiosk sits beside the exit, far from the waiting room. Patients know how to use it, but they keep walking past it, so the front desk ends up checking most of them in by hand. The kiosk software itself works perfectly."	Workflow mismatch

Groups: three targets labeled "Software defect," "Training gap," "Workflow mismatch," each with a "Place selected case here" button.

Feedback (shown after each placement; a wrong placement leaves the card in the list with a "Try again" prompt):

Card	Correct	Not quite
1	"Correct. Same failure, for everyone, everywhere: the printer queue is not doing what it was built to do."	"Not quite. Look at who is affected. When it fails the same way for everyone, the fault is usually in the software itself. Try again."
2	"Correct. The tool works, and the people who were shown it use it fine. The people who struggled are the ones nobody showed."	"Not quite. Compare the people who were shown the portal with the people who were not. That difference points to training. Try again."

3	"Correct. The software works and the patients know how to use it. Where the kiosk sits does not match how patients move through the clinic."	"Not quite. Notice that nothing is wrong with the software and nobody lacks skill. The setup does not fit the way the work actually happens. Try again."

A correct placement moves the card into its group and keeps its feedback beside it.

Completion message (all three placed correctly):

> You have sorted all three. In real calls the clues are rarely this tidy, which is why the next two moves matter: you find out by listening and by asking.

Build (web):

- Cards are focusable elements that can be chosen with Enter or Space, or dragged. A card is chosen, then a group's "Place selected case here" button is pressed (or the card is dropped on the group).
- When all three are correctly placed, the completion message shows and Next appears in the bottom bar.
- Not tracked (D-162).

---

4. Slide 3: Move one, acknowledge before you troubleshoot

Layout: a short scenario brief at top; two side-by-side cards beneath, each with a button; a teaching panel that appears when both reactions have been revealed.

On-screen text:

> # Move one: acknowledge before you troubleshoot
>
> An angry caller wants to know that someone has noticed what this is costing them. Here is the start of a call.
>
> **The caller:** Tomás, a warehouse supervisor.
> "*The new barcode scanners keep freezing in the middle of a shift. I have trucks waiting and my crew is standing around. Fix this.*"
>
> Two specialists answer. Select a button under each one to see what Tomás hears.
>
> **Specialist A**
> "That sounds like a real problem. Trucks are waiting and your crew is idle, and that is costing you. Tell me what your crew is doing at the moment a scanner freezes."
>
> **Specialist B**
> "I understand your frustration, but I am going to need a ticket number before I can do anything." (A **ticket** is the record of a problem in the support system.)

Interaction: each card has a "What Tomás hears" button that reveals the caller's reaction. The label says what the button reveals; the earlier label, "Hear it as Tomás", suggested audio was about to play (D-189).

Revealed text:

Card	Revealed text
A	"Someone gets it. They know what is at stake and they want to know what is happening. He starts to explain."
B	"A script. The sympathy is a lead-in to a demand, and now he has to prove his problem before anyone will listen."

The revealed text no longer begins "What Tomás hears:", since the button now says it (D-189).

Teaching panel (appears after both reactions have been revealed):

> **Acknowledgment helps when it leads somewhere.**
> Name two things: how the caller feels, and what is at stake for them. Then move into a real question or a real commitment.
>
> **It backfires when it sets up a "but."**
> A caller hears the "but" and forgets everything before it. That is why "I understand, but..." sounds like a technique rather than care.

Build (web):

- Specialist A and B text is on the page from the start, so it is readable without pressing anything; the button reveals the caller's reaction only.
- When both reactions have been revealed, the teaching panel shows and Next appears.

Design notes: the example call is a different industry and a different complaint from the simulation (D-151). Specialist A includes an open question after the acknowledgment on purpose: this is acknowledgment followed by forward motion, which is the pattern the simulation rewards. Specialist B is the acknowledge-and-pivot pattern the simulation treats as an escalating move (D-041).

---

5. Slide 4: Move two, ask to find the cause

Layout: two parts on one slide. The **explanation** comes first: the principle and a Start button. The **questions** come second: the caller's problem in the caller-quote style used on slide 3, kept above all three question pairs, then one pair at a time as radio choices with a Submit button. A "Review the explanation" button lets the learner go back from the questions to the explanation at any point (D-189).

On-screen text (explanation):

> # Move two: ask to find the cause
>
> You rarely know the cause at the start of a call. Questions are how you find it.
>
> **Closed questions** can be answered "yes" or "no." They confirm what you already suspect.
> **Open questions** invite a description. They bring out what you did not know to ask about.
>
> The most useful open questions are about **what people do and see at the moment the problem happens.**
>
> One more thing: **order matters.** A person who does not feel heard will not hand over detail. Acknowledge first, then ask.
>
> Now try three. Which question would reveal more?
>
> [Start]

On-screen text (questions, above every pair):

> **The caller:** "*The expense app keeps rejecting my team's receipts.*"
>
> [Review the explanation]

The caller's line stays visible above all three pairs so the learner does not have to hold it in memory from the previous screen (D-189).

The three pairs:

Pair	Question A	Question B	Correct
1	"Did you update the app recently?"	"Tell me what happens, step by step, when someone on your team submits a receipt."	B
2	"What is your team doing just before a receipt gets rejected?"	"Has your team been trained on the app?"	A
3	"Is this only happening on phones?"	"Where in the process do people tend to get stuck?"	B

Interaction: choose one question per pair (radio buttons), then Submit. Feedback follows each choice.

Feedback:

Pair	Correct	Not quite
1	"Correct. Question B asks for a description, so you hear the whole sequence. Question A can only be answered yes or no, and it points at one guess."	"Not quite. Question A confirms one idea you already had. Question B asks for the whole story, which is where surprises come from."
2	"Correct. Question A asks what people are doing right before things go wrong, which tells you where to look. Question B tests only one idea, and it is a closed question."	"Not quite. Question B is closed and tests a single assumption (that this is about training). Question A asks about what is actually happening."
3	"Correct. Question B asks where people get stuck without suggesting an answer. Question A narrows the problem before you know enough to narrow it."	"Not quite. Question A narrows the search to one guess. Question B lets the caller show you where the trouble is."

Completion message (after pair 3 is answered, shown in the questions part):

> **Good questions describe the moment, not the guess.**

Build (web):

- A first wrong answer shows the "Not quite" feedback followed by "Try again." and leaves the question open. A second wrong answer shows the feedback, then says "The stronger choice was Question X." followed by the correct-answer explanation, and the pair closes.
- After a pair closes, a Continue button moves to the next pair. After the third, the completion message shows and Next appears.
- "Review the explanation" returns to the explanation part. While the learner is there, the Start button reads "Back to the questions" and returns to the same pair, with progress and any completed state kept (D-189).
- The completion message sits inside the questions part, so it does not appear under the explanation during a review (D-189).
- The stylesheet adds a space after the bold terms "Closed questions" and "Open questions", which otherwise ran into the definition that follows them (D-189).
- Correct answers alternate positions across pairs (B, A, B) so position is not a clue.
- Not tracked (D-162).

Design notes: "Order matters" is taught here in the same plain terms the performer enforces: Marcus withholds detail from someone who has not shown they heard him (D-040). The slide does not describe the mechanism, only the principle.

---

6. Slide 5: Move three, say what you found, route it, commit to what you control

Layout: a short intro, an ordering activity with three numbered slots, then a contrast panel.

On-screen text (intro):

> # Move three: say what you found, route it, commit to what you control

> Once you understand the problem, three things bring the call to a close.
>
> 1. **Say it back.** Tell the caller what you think the cause is, and check that you have it right.
> 2. **Route it.** If you cannot fix it yourself, pass it to someone who can. That is called **escalating**: passing a problem to the person with the authority or skills to fix it. For software, that is often the **product owner**, the person responsible for deciding how a piece of software works and what changes are made to it.
> 3. **Commit.** Tell the caller **who** will follow up and **when**, without waiting to be asked.
>
> Put the three lines below in the order a good call follows them.
>
> [Start]

Ordering activity. After Start, the instruction reads: "Pick a line, then choose its place in the order. (You can also drag a line into a slot.)" The three cards start in a mixed order. A "Check order" button becomes available when all three slots are filled.

Correct position	Card text
1	"Here is what I am hearing: the scanners freeze whenever your crew scans a damaged label. Have I got that right?"
2	"That is something the scanner vendor's support lead can change, so I am sending it to them today with exactly what you described."
3	"I will call you Thursday at 10 to tell you where it stands."

Feedback:

> **Correct.** Say it back and get a yes, route it to the person who can fix it, then tell the caller who will follow up and when.

> **Not quite.** Check the order again. You cannot route a problem you have not confirmed, and you cannot promise a follow-up until you know who owns it.

Contrast panel (revealed after the ordering is correct):

> **Two promises, one you can keep**
>
> **Within your control:** "I will call you Thursday with an update."
> You decide whether you make that call.
>
> **Not within your control:** "I will have this fixed by Thursday."
> Someone else decides when it is fixed.
>
> **Promise what you control.**
>
> **And avoid two shortcuts that callers have often heard before:** "It is working as designed" and "That is a training issue." They are conclusions, not diagnoses. Said before you understand the problem, they sound like a brush-off.

Build (web):

- Choose a line, then choose a slot (or drag it into a slot). Choosing a filled slot with no line selected returns its line to the list, so an order can be corrected.
- A wrong order leaves the arrangement in place with the "Not quite" message, so the learner can adjust it.
- A correct order locks the activity, reveals the contrast panel, and shows Next.
- The contrast panel is static text (D-170), not a second interaction.
- Not tracked (D-162).

Design notes: the "commit to what you control" teaching point mirrors the evaluator's distinction between committing to communication and committing to resolution, so a learner is not penalized for doing what the training seemed to ask (D-051, D-154). The sentence about volunteering the owner and the day matches D-129. No point values or rubric language appear anywhere on the slide (D-153). The two shortcuts named in the last paragraph correspond to the simulation's worst-move trigger and its training-gap trigger, taught here as a principle rather than as a list of forbidden phrases.

---

7. Slide 6: Before you pick up

Layout: a short briefing on the left, "how the call works" on the right, a Start the call link. This slide must make sense to a viewer who has seen none of slides 2 to 5, because the skip button lands here.

On-screen text:

> # Before you pick up
>
> *New here? This call is the practice activity from a short course on handling frustrated callers.*
>
> **The caller**
> Marcus is a **regional sales lead**: a manager who oversees a team of sales representatives (his **reps**) in one part of the country. It has been a bad month for his team. He uses the company's **CRM** (**customer relationship management** system), the software a sales team uses to keep track of potential customers and how close each one is to buying. He is calling **IT** (**information technology**) about it. He is already upset.
>
> **Your role**
> You are the tech support specialist who has picked up the phone.
>
> **How the call works**
>
> - You type, and Marcus replies, like a live call shown as a running transcript.
> - There is no checklist on screen and no timer, just as on a real call. A real caller's patience is limited, though, and so is this call.
> - When the call ends, you get a score and detailed feedback.
> - You can try again.
> - You can invent names and dates, such as the name of the person you would pass a problem to. Keep dates realistic.
>
> [Start the call]

Interaction: one link.

Build (web):

- "Start the call" is a link to the simulation's production address with `?embed=1` (D-190), set in one place: the `data-sim-url` attribute on the page's `<body>` tag. It opens in the same tab. The parameter makes the simulation skip its own landing view and start the call, because this slide has already set the scene. A visitor who opens the simulation directly, without the parameter, still gets the landing view.
- Nothing is passed to the simulation, and nothing comes back.
- Slide 6 states the caller's job title and tool but says nothing about what has changed or gone wrong in that tool.

Design notes: this slide names the CRM, which the outline had said to leave out (D-166). Naming it is safe because the simulation's first line names it anyway; what stays out is anything about the problem itself. Because the link suppresses the simulation's landing view, this slide and the glossary are where a learner arriving from the training meets the definition (D-166, and see open item 2).

---

8. Debrief slide: not built

The v0.1 storyboard drafted an optional debrief after the simulation: a score line, a pass or fail branch, a reflection prompt, and a retry button. It read the learner's score and pass state from the simulation, which two separate sites do not share, so it is not built (D-183). The simulation's own results view carries the score and detailed feedback.

The draft copy is kept here as a record, in case a short "After the call" page is added later:

> **Pass:** You handled the call well. Before you move on, think about one moment in the call where you could have asked a better question. What would you have asked?
>
> **Not yet:** This call did not reach the standard yet. That is what practice calls are for. Read your feedback, then try again with one change in mind.

9. Glossary dialog (opened from the persistent Glossary button)

> # Glossary
>
> **CRM (customer relationship management system).** Software a sales team uses to track potential customers and how close each one is to buying.
>
> **COO (chief operating officer).** A senior executive responsible for running a company's day-to-day operations.
>
> **Defect.** See *Software defect*.
>
> **Escalate.** To pass a problem to someone with the authority or skills to fix it.
>
> **HR (human resources).** The department that handles hiring, benefits, and employee records.
>
> **IT (information technology).** The department and the systems that keep a company's computers, software, and networks running. Also called tech support or IT support.
>
> **Product owner.** The person responsible for deciding how a piece of software works and what changes are made to it.
>
> **Regional sales lead.** A manager who oversees a team of sales representatives in one part of the country.
>
> **Rep (sales representative).** A salesperson.
>
> **Software defect.** The software is not doing what it was built to do.
>
> **Symptom.** What someone notices, as opposed to why it is happening.
>
> **Ticket.** The record of a problem in the support system.
>
> **Training gap.** The software works and fits the work, but people were never shown how to use it.
>
> **Troubleshooting.** Working through possible causes, one at a time, to find and fix a problem.
>
> **VP (vice president).** A senior manager, above department heads.
>
> **Workflow.** The series of steps a person follows to get a piece of work done.
>
> **Workflow mismatch.** The software works as built, but it does not fit how the work actually happens.

Build (web): a dialog with a Close button, alphabetized and scrollable. It closes with the Escape key or a click outside it. The link on slide 1 opens it at the matching term, which is highlighted. COO and VP are included because Marcus may use them in the simulation, which does not define them.

10. Acronym and jargon audit

Every acronym and piece of jargon in this storyboard, where it first appears on each slide, and where it is defined.

Term	Slides where it appears	Defined in line on	Glossary
IT (information technology)	1, 6	1, 6	Yes
CRM (customer relationship management)	6	6	Yes
HR (human resources)	2	2	Yes
VP, COO	glossary only; used by Marcus in the simulation	Glossary	Yes
Ticket	3	3	Yes
Escalate / escalating	5	5	Yes
Product owner	5	5	Yes
Workflow	2	2	Yes
Software defect, training gap, workflow mismatch	1, 2	2	Yes
Troubleshooting	1, 3	Glossary (linked from slide 1)	Yes
Regional sales lead, rep	6	6	Yes
Symptom, cause	2	2	Yes

11. Open items

1. **Confirm that naming the CRM on slide 6 is acceptable.** It reverses the outline's original "no mention of the CRM." It is safe because the simulation's first line already names it, and slide 6 says nothing about what is wrong with it (D-166). Still open.
2. **The simulation does not define CRM, COO, or VP.** Because the "Start the call" link uses `?embed=1` (D-190), a learner arriving from the training skips the simulation's landing view, so slide 6 and the glossary carry those definitions. If you ever want the application to carry its own definitions, that is an application change and is not proposed or built here.
3. **Debrief:** not built (D-183); the draft copy is kept in section 8.

Closed since v0.1: the story size question no longer applies (D-186). Closed since v0.2: whether the link carries `?embed=1` (D-190).

12. Changes from earlier versions

v0.1 to v0.2 (November 11, 2026)

Where	v0.1	v0.2
Slide 2, part C instruction	"Drag each case to the kind of cause it points to."	"Pick a case, then choose the kind of cause it points to. (You can also drag a case onto a group.)"
Slide 2, wrong placement	Card snaps back; feedback ends at the clue	Card stays in the list; "Try again." is added to the feedback
Slide 3 instruction	"Click each one to hear it the way Tomás does."	"Select each one to hear it the way Tomás does." The button on each card reads "Hear it as Tomás"
Slide 4, feedback after a second wrong answer	One retry, then "show the explanation"	The stronger choice is named, then the correct-answer explanation is shown
Slide 5, instruction	"Put the three lines below in the order a good call follows them" (drag only)	Same line, then: "Pick a line, then choose its place in the order. (You can also drag a line into a slot.)" A "Check order" button is added
Slide 6, button	"Start the call" jumped to a simulation	"Start the call" is a link to the simulation's address

	slide	
Section 8	Debrief slide drafted	Not built (D-183); copy kept as a record
Section 9	One master layer	A dialog; the link on slide 1 opens it at the term
Global	Story size, slide master, triggers, variables	Responsive layout, a fixed navigation bar, no stored state

v0.2 to v0.3 (November 12, 2026; D-189, D-190)

Where	v0.2	v0.3
Slide 3 instruction	"Select each one to hear it the way Tomás does."	"Select a button under each one to see what Tomás hears."
Slide 3 buttons	"Hear it as Tomás"	"What Tomás hears"
Slide 3 revealed text	Began "What Tomás hears:"	Prefix removed, since the button now says it
Slide 4 caller quote	Shown once at the end of the setup text	Kept above all three question pairs, in the slide 3 caller-quote style
Slide 4 navigation	No way back to the explanation once the questions began	"Review the explanation" returns to it; the Start button then reads "Back to the questions" and keeps progress
Slide 4 completion message	Placed at the end of the slide	Inside the questions part, so it does not show during a review
Slide 4 defined terms	"Closed questions" and "Open questions" ran into their definitions on screen	A space is added after each term
Slide 6 link	Plain production address; `?embed=1` open (D-182)	Production address with `?embed=1` (D-190)